

Experience Using Adjuvant Bacteriophage Therapy for the Treatment of 10 Recalcitrant Periprosthetic Joint Infections: A Case Series

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Periprosthetic joint infections are a devastating complication of joint replacement surgery. One novel therapeutic that has potential to change the current treatment paradigm is bacteriophage therapy. Herein, we discuss our experiences with bacteriophage therapy for 10 recalcitrant periprosthetic joint infections and review the treatment protocols utilized to achieve successful outcomes.

Keywords. bacteriophage therapy; periprosthetic joint infections; *Staphylococcus*; intraarticular; debridement and implant retention surgery.

Periprosthetic joint infections (PJIs) are associated with debilitating morbidity, increased mortality risk, and immense financial implications [1, 2]. In part, this is because conventional antibiotics have limited activity against metabolically reduced bacterial phenotypes [3, 4]. Consequently, novel agents are needed to change the current treatment paradigm. Bacteriophage therapy is one such agent, given bacteriophages have evolved with bacteria to possess innate abilities to degrade biofilm matrices and lyse bacteria in metabolically reduced phenotypes [5]. Several isolated case reports have documented the successful treatment of recalcitrant PJI with bacteriophage therapy, but the reproducibility of outcomes is unproven [6]. Herein, we present our experiences treating 10 recalcitrant PJIs with adjuvant bacteriophage therapy and discuss the protocols used to create effective and reproducible treatment outcomes.

METHODS

Between September 2019 and February 2022, 10 patients with chronic recalcitrant knee and/or hip PJI were treated with adjuvant bacteriophage therapy at the University of Maryland Medical System. These patients were treated under the US Food and Drug Administration's (FDA) compassionate use expanded access pathway. All patients had failed previous conventional surgical and antibiotic therapies, resulting in most to have been recommended for arthrodesis or amputation. Adjuvant bacteriophage therapy was used with either debridement and implant retention (DAIR, n = 6) or revision surgery (1- or 2-stage, n = 4). Retention of the prosthesis was attempted in all cases. However, this was not feasible in 4 cases, but in these cases chronic infection niduses were left in situ: 2 patients had additional chronically infected hardware left in situ, while another 2 patients had extensive chronic osteomyelitis left in situ.

In all of the cases, bacteria cultured from an arthrocentesis were used to select for a bacteriophage therapy. This selection was conducted by ensuring plaque formation on agar plates with the clinical bacterial isolate and bacteriophage-induced bacterial growth inhibition assays. Bacteriophage activity against the bacterial clinical isolates in different phenotypic states was also conducted to ensure activity against biofilms. All but 2 bacteriophage therapeutics used in this cohort were created by the Yale Center for Phage Biology and Therapy (New Haven, Connecticut). The other 2 bacteriophage therapeutics were created by Adaptive Phage Therapeutics (Gaithersburg, Maryland).

Figure 1A discusses the 2 protocols used to treat these recalcitrant PJI. Each patient underwent discussions about theoretical risks and potential benefits associated with each protocol and subsequently each treatment was individualized based on patient and clinician preferences. Every patient was administered intraarticular bacteriophage therapy at the end of surgery, except patient 1 who had intraarticular bacteriophage therapy administered with an arthrocentesis 6 weeks after DAIR. Additional bacteriophage therapy was also administered on postoperative day 1, either with repeat daily intraarticular doses and/or daily intravenous (IV) bacteriophage therapy. Repeat intraarticular doses were administered through Hickman catheters placed at time of surgery. Intraarticular doses were diluted in 20 mL of normal saline whereas IV doses were diluted in 50 mL of normal saline and infused over 30 minutes. Standard-of-care IV antibiotic therapy was administered to all patients for 6 weeks, followed by oral suppression therapy in patients who underwent DAIR, except for patient 6, who had significant antibiotic intolerances and could not tolerate prolonged suppression therapy [4].

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A Protocols for Bacteriophage Therapy in PJI		B					
Protocol 1: - Intraoperative bacteriophage therapy¹: - Then - Intravenous bacteriophage therapy²: 1 dose daily for 5 days Protocol 2: - Intraoperative bacteriophage therapy¹: - Plus - Placement of hickman Catheters³ - Then - Intraarticular bacteriophage Therapy⁴: 1 dose daily for 4 days - Plus - Intravenous bacteriophage therapy²: 1 dose daily for 5 days Both protocols use standard of care antibiotic therapy ¹Doses given at the end of surgery are diluted in normal saline to create 20 mL solutions. ²Intravenous doses are diluted in 50 mL of normal saline and infused over 30 minutes. ³Hickman catheters are removed after last dose of intraarticular bacteriophage therapy. ⁴Intraarticular doses are diluted in 20 mL of normal saline.		Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6
Bacteria		<i>Enterococcus faecalis</i>	<i>Klebsiella pneumoniae</i>	<i>Staphylococcus epidermidis</i>	<i>Staphylococcus lugdunensis</i>	MRSA	MRSA
Location		Knee	Shoulder	Knee	Knee	Knee	Hip and knee
Sinus tract		Sinus tract	Sinus tract	None	Sinus tract	Sinus tract	None
Bacteriophage		EF 1	KP 1 & 2	PM448	Mallokai	Mallokai	SaWIC0488e1
Duration of daily IA phage therapy		1 day	4 days	1 day	4 days	4 days	1 day
Duration of daily IV phage therapy		5 days	2 days	None	5 days	2 days	4 days
Titers (PFU/mL)		1 × 10 ¹⁰	1 × 10 ¹⁰	1 × 10 ¹⁰	1 × 10 ¹⁰	1 × 10 ¹⁰	1 × 10 ¹⁰
Adverse events		None	None	Elevated LFT	None	Elevated LFT	Elevated LFT
Outcome		No clinical recurrence 1.2 years	No clinical recurrence 1.2 years	No clinical recurrence but arthrodesis secondary to infected port seeding joint	No clinical recurrence 5 months	No clinical recurrence 5 months	No clinical recurrence 1.5 years

C	Bacteria	Location	Sinus tract	Surgical procedure	Residual hardware/chronic nidus of infection	Bacteriophage	Duration of daily IA phage therapy	Duration of daily IV phage therapy	Titers (PFU/mL)	Adverse events	Outcome
Patient 7	MRSA	Knee	None	1-stage revision	Pelvic trauma hardware	Mallokai	4 days	None	1 × 10 ¹⁰	None	No clinical recurrence 1.2 years
Patient 8	MRSA	Hip and knee	Sinus tract	2-stage revision	Extensive chronic osteomyelitis	Mallokai	1 day	3 days	1 × 10 ¹⁰	Elevated LFT	No clinical recurrence 1.2 years
Patient 9	MRSA	Knee	Sinus tract	2-stage revision	Extensive chronic osteomyelitis	SaGR51Φ1	1 day	3 days	2.4 × 10 ⁹	Elevated LFT	No clinical recurrence 2.5 years
Patient 10	<i>Staphylococcus lugdunensis</i>	Knee	None	2-stage revision	Residual hardware	Mallokai	4 days	5 days	1 × 10 ¹⁰	None	No clinical recurrence but amputation secondary soft tissue contractures and exposed prosthetic

Figure 1. Protocols and description of recalcitrant periprosthetic joint infection (PJI) cases treated with bacteriophage therapy. **A**, Two protocols used to treat recalcitrant PJI at the University of Maryland Medical System. **B**, Six patients with recalcitrant PJI treated with bacteriophage therapy and debridement and implant retention. **C**, Four patients with recalcitrant PJI treated with bacteriophage therapy and revision surgery. Abbreviations: IA, intraarticular; IV, intravenous; LFT, liver function test; MRSA, methicillin-resistant *Staphylococcus aureus*; PFU, plaque-forming units; PJI, periprosthetic joint infection.

RESULTS

Seven patients (60%) had sinus tract infections and 5 patients (50%) had methicillin-resistant *Staphylococcus aureus* infections (Figure 1B and 1C). Bacteriophage therapy was well tolerated in all patients. Two patients (4 and 10) had IV doses infiltrate soft tissues of the upper extremities with no detrimental side effects. Reversible transaminitis occurred in 5 patients, slightly truncating treatment duration given guidelines created in collaboration with the FDA. In all cases, transaminitis resolved after cessation of bacteriophage therapy and no permanent liver damage occurred.

No patient has had recurrence of their PJI with the infecting organism that had caused their recalcitrant PJI. Patient 3 had a new infection originate from a chronically implanted port, requiring arthrodesis and port removal. Patient 10 ultimately required amputation given extensive soft tissue contractures from rheumatoid arthritis that prevented wound healing and resulted in an exposed prosthesis. However, at the time of amputation, there was no pathological evidence of osteomyelitis, and cultures had no microbial growth.

DISCUSSION

To our knowledge, this is the largest cohort of recalcitrant PJI successfully treated with adjuvant bacteriophage therapy to reduce morbidity and achieve limb salvage. While these cases undoubtedly require further longer-term follow-up, the success seen in these recalcitrant PJI cases supports the safety and effectiveness of bacteriophage therapy when used as an adjuvant with conventional surgical and medical treatments. Using bacteriophages as adjuvants and not as stand-alone therapeutics does reduce the appeal of this therapeutic, but there are benefits of adjuvant usage [7]. Furthermore, the use of suppression antibiotics with DAIR is recommended for acute PJI, but DAIR is not recommended for chronic PJI given high failure rates [4]. Therefore, the successes seen here should reinforce the potential of adjuvant bacteriophage therapy even though prolonged suppression antibiotics were used in 5 of the 6 DAIR cases and especially since these were recalcitrant PJI cases. Moreover, when bacteriophage therapy is used as an adjuvant to standard surgical and medical interventions, treatment protocols can be standardized. This allows for the establishment of

reproducible outcomes and a foundation to conduct clinical trials to prove bacteriophage effectiveness. Consequently, if successful outcomes continue to be proven, treatment of chronic PJI may be feasible with prosthesis retention, which would revolutionize this field.

The 2 bacteriophage therapy protocols utilized to achieve these outcomes are discussed in Figure 1A. These protocols use short courses of bacteriophage therapy to mitigate the resistance and neutralizing antibody responses that can occur with prolonged durations of bacteriophage therapies [6]. Protocol 1 uses an intraoperative dose and short courses of IV bacteriophage therapy to reach chronic infection niduses not accessible with intraoperative administration [7]. Protocol 2 also utilizes additional intraarticular doses with Hickman catheters. This allows for repeated direct administration of bacteriophage therapy to achieve high multiplicities of infection with each dose. However, these catheters are controversial in PJI, given the concern for retrograde translocation of skin flora into the joint [8]. We mitigated these negative consequences by removing the catheters after 4 days of therapy. While these protocols are attractive, bacteriophage pharmacokinetic research is needed to better clarify the routes discussed here.

In this cohort, all bacteriophage doses were well tolerated, but reversible transaminitis limited the ability to finish all planned doses for some patients. Patient 9 had the most significant transaminitis, in which his aspartate aminotransferase increased to 700 [9]. The exact mechanism of the transaminitis is unknown, but it does not appear to be a direct consequence of the bacteriophages themselves given the lack of adverse events seen by others [6]. Rather, the degradation of the biofilm may release antigens that stimulate cytokine responses priming hepatocytes to become inflamed when continuous administered bacteriophages need to be hepatically cleared [10]. However, more research is needed to better clarify these findings with respect to cytokine responses.

Correlated with cytokine responses, we ensured that all bacteriophage therapeutics had levels of endotoxin units <1 EU/mL and sterility was proven with third-party testing. All bacteriophage therapeutics also lacked integrase genes and were considered obligate lytic phages. However, 2 patients (7 and 8) had low levels of staphylococcal enterotoxin A in their therapeutic since their bacteriophages were propagated on their *Staphylococcus aureus* clinical isolate [11]. This caused patient 7 to not receive IV bacteriophage therapy after discussion with the FDA. In addition, patient 9 had a prophage incorporated with his therapeutic that was isolated retrospectively. These findings reinforce that clinicians should be cognizant about other contaminants beyond endotoxins that may be present in these therapeutics [11].

While the outcomes and tolerability are encouraging, not all bacteriophages have the same potential to be PJI therapeutics. Consequently, we only used bacteriophages that formed

plaques on agar plates and had growth inhibition times that were >24 hours. Moreover, only bacteriophages that had activity to planktonic and biofilm phenotypes of the patients' clinical isolates were used. Unfortunately, no standardized growth inhibition time has been determined, but assessing PJI bacteriophage therapeutics for prolonged growth inhibition and biofilm activity is likely paramount to ensure activity across all PJI phenotypes [12].

In conclusion, this brief report demonstrates the safety and effectiveness of bacteriophage therapy when used as adjuvants to surgical and medical PJI treatments. It proves that bacteriophage therapy should be further evaluated to reduce the morbidity, mortality, and enormous financial ramifications associated with PJI. However, before definitive clinical trials are conducted, additional research is needed to clarify several aspects of this therapeutic to thereby create generalizable, reproducible, and effective outcomes.

Notes

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Potential conflicts of interest. J. B. D. has a patent pending with respect to the use of bacteriophage therapy in periprosthetic joint infections, including consulting fees from Adaptive Phage Therapeutics, but has stopped receiving these fees as of February 2022. B. C. reports grants or contracts from Emily's Entourage; support for attending meetings and/or travel from Yale University; and data and safety monitoring board membership for a phase 1b/2 clinical trial (Division of Microbiology and Infectious Diseases, National Institute of Allergy and Infectious Diseases). T. M. reports royalties from Globus Medical; consulting fees from Stryker Medical and Depuy-Synthes; and a leadership or fiduciary role on the University of Maryland Joint Committee. All other authors report no potential conflicts.

All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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